

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. (CL) Examination

November 2013

CL203.01 Fluid Mechanics - I

Date: 26.11.2013, Tuesday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) State Newton's Law of viscosity and explain Newtonian fluids and Non-Newtonian fluids. [03]
- (b) The space between two square flat parallel plates is filled with oil. Each side of the plate is 720 mm. The thickness of the oil film is 15 mm. The upper plate, which moves at 3 m/s requires a force of 120 N to maintain the speed. Determine (i) The dynamic viscosity of the oil (ii) The kinematic viscosity of oil if the specific gravity of oil is 0.95. [04]
- Q - 2 (a) Enlist different types of fluid flow and define any two. [04]
- (b) For the following stream function calculate velocity at a point (1, 2): [05]
- (i) $\psi = 3xy$ (ii) $\psi = 3x^2y - y^3$.
- (c) A vertical sharp-edged orifice 120 mm in diameter is discharging water at the rate of 98.2 lit/sec. under a constant head of 10 m. A point on the jet, measured from the vena contracta of the jet has co-ordinates 4.5 m horizontal and 0.54 m vertical. Find the following for the orifice. [05]
- (i) Co-efficient of velocity (ii) Co-efficient of discharge, and (iii) Co-efficient of contraction.

OR

- Q - 2 (a) State Continuity equation. [04]
- The diameters of a pipe at the section 1-1 and 2-2 are 200 mm and 300 mm respectively. If the velocity of water flowing through the pipe at section 1-1 is 4 m/s, find (i) discharge through the pipe, and (ii) Velocity of water at section 2-2.
- (b) Explain the Lagrangian method for motion of fluid particles. [05]
- (c) Define Notch and Weir. Also give the classification of both. [05]
- Q - 3 (a) Derive an expression for discharge over a triangular notch. [05]

- (b) Derive an expression for discharge through a large rectangular orifice. [04]

OR

- (b) A horizontal venturimeter with inlet diameter 200 mm and throat diameter 100 mm is employed to measure the flow of water. The reading of the differential manometer connected to the inlet is 180 mm of mercury. If the co-efficient of discharge is 0.98, determine the rate of flow. [04]
- (c) Define Vortex flow. Also enlist types of vortex flow. [05]
Find the maximum speed of an open circular cylinder, having 180 mm diameter, 1200 mm length and containing water up to a height of 960 mm, at which it should be rotated about its vertical axis so that no water spills.

OR

- (c) Derive continuity equation for 3-D flow in Cartesian co-ordinates. [05]

SECTION – II

- Q – 4 (a) Enumerate the practical application of Bernoulli's equation. [02]
- (b) Water flows in a circular pipe. At section 1-1 the diameter is 0.3 m, the static pressure is 260 kPa, the velocity is 3 m/s and the elevation is 10 m above ground level. The elevation at a section 2-2 is 0 m, and the pipe diameter is 0.15 m. Find out the pressure at the section 2-2. Neglect the friction effect. Assume density of water to be 999 kg/m^3 . [05]
- Q - 5 (a) Define Hydraulic gradient line and Compound pipe. [02]
- (b) A compound piping system consists of 1800 m of 50 cm diameter, 1200 m of 40 cm diameter and 600 m of 30 cm diameter pipes of the same material connected in series. What is the equivalent length of a 40 cm diameter pipe of the same material? Also state the assumptions made. [03]
- (c) Define Atmospheric pressure, Gauge pressure, Vacuum pressure and Absolute pressure. [04]
- (d) In a pipe of diameter 350 mm and length 75 m water is flowing at a velocity of 2.8 m/s. Find the head loss due to friction using (i) Darcy-Weisbach formula and (ii) Chezy's formula for which $C = 55$. Assume kinematic viscosity of water as 0.012 stoke. [05]

OR

- Q - 5 (a) A rectangular plate 3 m long and 1 m wide is immersed vertically in water such a way that its 3 m side is parallel to the water surface and is 1 m below it. Find (i) Total pressure on the plate, and (ii) Position of centre of pressure. [04]

- (b) Derive an expression for total head loss when three pipes of different lengths and different diameters are connected in series to make a compound pipe. The end of this pipe are connected with two reservoirs whose difference of water level is H . Assume co-efficient of friction for these pipes is same and minor losses are negligible. [05]
- (c) The diameter of a horizontal pipe which is 300 mm is suddenly enlarged to 600 mm. [05]
The rate of flow of water through this pipe is $0.4 \text{ m}^3/\text{sec}$. If the intensity of pressure in the smaller pipe is 125 kN/m^2 , determine (i) Loss of head, due to sudden enlargement, (ii) Intensity of pressure in the large pipe.

Q - 6 (a) Explain momentum correction factor. [02]

- (b) Water is flowing through a pipe having diameters 600 mm and 400 mm at the bottom and upper end respectively. The intensity of pressure at the bottom end is 350 kN/m^2 and the pressure at the upper end is 100 kN/m^2 . Determine the difference in datum head if the rate of flow through the pipe is 60 litres/sec. [04]

(c) Explain equilibrium conditions of floating body with neat sketch. [04]

(d) Define manometer. [04]

A U-tube manometer is used to measure the pressure of oil of specific gravity 0.85 flowing in a pipe line. Its left end is connected to the pipe and the right – limb is open to atmosphere. The centre of the pipe is 100 mm below the level of mercury (Specific gravity 13.6) in the right limb. If the difference of mercury in the two limbs is 160 mm, determine the pressure of the oil in the pipe.

OR

Q - 6 (a) Derive the expression for total pressure force on a vertically submerged surface. [04]

- (b) A wooden block of specific gravity 0.75 floats in water. If the size of the block is 1 m * 0.5 m * 0.4 m, find its Metacentric height. [04]

(c) State Euler's equation of motion. Describe the forces considered in the same. Also [06]
derive the Bernoulli's equation from Euler's equation.
